Year 8
Mathematics

Unit 3: Working with Powers

3.5 Substituting and Solving

Substituting and Solving

STARTER · 5 MIN

Date: _____ **Title: Substituting and Solving**

LEARNING OBJECTIVES

- To be able to substitute into expressions and construct and solve equations

STARTER *Answer in your book. No calculators.*

1. Perimeter of a $5x + 2$ by $2x + 1$ rectangle?
2. Find that perimeter when $x = 2$
3. Solve $3x + 4 = 19$
4. Work out $2 \times (-3)^2$

Starter — Answers

STARTER

Tick or correct in green pen

1. $14x + 6$

2. 34 cm

3. $x = 5$

4. 18

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can substitute given values into algebraic expressions correctly	☐
SILVER	I can construct simple equations from worded or geometric problems	☐
GOLD	I can solve equations to find the value of an unknown	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Substituting and Solving

INTRODUCTION · 3 MIN

Substitute: replace each letter with its value. Solve: find the value of the unknown.

Substitute

Use brackets for negatives.

$x = -3$: $x^2 = (-3)^2 = 9$

Linear expression

The highest power is 1.

$2x + 5$ and $a + 2b$

Construct

Write an equation from a problem.

Perimeter = 50: $18a + 2b = 50$

Solve

Use inverse operations.

$96 - 3x = 81 \rightarrow x = 5$

Examples 1 and 2 — Substituting Values

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint Negatives in brackets Then BIDMAS

Example 1

$m = 4$ and $x = -3$

Find the value of

$2m^2 + mx$

Example 2

$x = 2$, $y = -3$, $z = 5$

Find the value of

$z(x + y)$

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Substitute, then BIDMAS

$$\begin{aligned} 2m^2 + mx &= 2 \times 4^2 + 4 \times (-3) \\ &= 32 - 12 \\ &= \mathbf{20} \end{aligned}$$

Tip: square first, then × 2.

Example 2

Brackets first

$$\begin{aligned} z(x + y) &= 5 \times (2 + -3) \\ &= 5 \times (-1) \\ &= \mathbf{-5} \end{aligned}$$

Tip: 2 + −3 = −1.

Task 1

PRACTICE · 9 MIN

BRONZE

x = 2, y = −3, z = 5. Find the value

Hint Negatives in brackets Then BIDMAS

1. $x + y + z$
2. $3x - y$
3. $2x(y + z) + 4$
4. $x^2 + y^2 + z^2$
5. $6x - 3(y - z)$

★ xyz

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. **4**
2. **9**
3. **12**
4. **38**
5. **36**
★ **−30**

Examples 3 and 4 — Roots and Formulae

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint Work out under the root first Expand, then solve

Example 3

$m = 4$ and $x = -3$

Find the value of

$\sqrt{x^2 + m^2}$

Example 4

$s = ut + 5t^2$

Find s when $u = 10$ and $t = 4$.

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Work out under the root first

$x^2 + m^2 = 9 + 16$

$= 25$

$\sqrt{25} = 5$

Tip: $(-3)^2 = 9$.

Example 4

Substitute, then BIDMAS

$s = 10 \times 4 + 5 \times 4^2$

$= 40 + 80$

$= 120$ m

Tip: $5 \times 16 = 80$.

Task 2

PRACTICE · 9 MIN

SILVER *Substitute*

Hint Work out under the root first Expand, then solve

1 $s = ut + 5t^2$

$u = 0,\ t = 2$

2 $s = ut + 5t^2$

$u = 4,\ t = 3$

3 $A = 8x^2 + 32x$

$x = 1$

4 $A = 8x^2 + 32x$

$x = 5$

5 $\sqrt{a^2 + b^2}$

$a = 6,\ b = 8$

★ $2(m + 4) + 3x$

$m = 4,\ x = -3$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. 20
2. 57
3. 40
4. 360
5. 10
★ 7

Examples 5 and 6 — Constructing and Solving Equations

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint Write the equation first Check your answer

Example 5

(3 marks)

A and B have equal areas.

A is 5 by $(x + 4)$.

B has area $86 - 6x$.

Find x.

Example 6

(3 marks)

A rectangle is $(2x + 3)$ by 4.

Its perimeter is 30 cm.

Find x.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Expand, then letters to one side

$5x + 20 = 86 - 6x$

$11x = 66$

$\boldsymbol{x = 6}$

Tip: add $6x$ to both sides.

Example 6

Write the perimeter, then solve

$2(2x + 3) + 8 = 30$

$4x + 14 = 30$

$\boldsymbol{x = 4}$

Tip: check: 11 by 4 gives 30.

Task 3

PRACTICE · 9 MIN

GOLD *Exam-style: construct and solve*

Hint Write the equation first Check your answer

1. Solve $4(x - 2) = 28$

2. Solve $96 - 3x = 81$

3. Solve $24y + 8 = 119 - 13y$

4. Areas 81 and $96 - 3x$ are equal. Find x.

5. Triangle sides x, $x + 2$, $2x$. Perimeter 22. Find x.

★ A card 4 by $(x + 5)$ has a 2 by 8 hole. Area left 24. Find x.

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. $x = 9$

2. $x = 5$

3. $y = 3$

4. $x = 5$

5. $x = 5$

★ $x = 5$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE $x = 2$, $y = -3$, $z = 5$: find $z(x + y)$

SILVER $x = 2$, $y = -3$, $z = 5$: find $x^2 + y^2 + z^2$

GOLD Solve $4(x - 2) = 28$

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE -5

SILVER 38

GOLD

$$x - 2 = 7$$

$$x = 9$$

Exam Practice — Substitution

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint Negatives in brackets Then BIDMAS

Question 1

(2 marks)

$m = 4$ and $x = -3$

Find the value of $2m^2 + mx$

.....

Question 2

(2 marks)

$m = 4$ and $x = -3$

Find the value of $2(x + 4) + 3m$

.....

Exam Practice — Roots and Solving

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint Work out under the root first
Expand, then solve

Question 3

(2 marks)

$m = 4$ and $x = -3$

Find the value of

$\sqrt{x^2 + m^2}$

.....

Question 4

(3 marks)

Solve $4(x - 2) = 28$

$x =$ .....

Exam Practice — Constructing Equations

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint Write the equation first Check your answer

Question 5

(3 marks)

A square has sides of $(2x + 3)$ cm.
Its perimeter is 44 cm.

Write an equation and find x.

$x =$ .....

Question 6

(3 marks)

A rectangle is $(x + 7)$ cm by 3 cm.
A 6 cm square has the same area.

Find x.

$x =$ .....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

$2 \times 16 + 4 \times (-3) = 32 - 12$ **M1**

$= \mathbf{20}$ **A1**

Tip: brackets round the negative.

Question 2

$2 \times 1 + 12 = 2 + 12$ **M1**

$= \mathbf{14}$ **A1**

Tip: −3 + 4 = 1.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

$9 + 16$	$= 25$	M1
$\sqrt{25}$	$= 5$	A1

Tip: $(-3)^2 = 9$.

Question 4

$4x - 8$	$= 28$	M1
$4x$	$= 36$	M1
x	$= 9$	A1

Tip: or divide by 4 first.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$4(2x + 3) = 44$ — M1

$8x = 32$ — M1

$\boldsymbol{x = 4}$ — A1

Tip: a square has 4 equal sides.

Question 6

$3(x + 7) = 36$ — M1

$x + 7 = 12$ — M1

$\boldsymbol{x = 5}$ — A1

Tip: square area = 6 × 6.